Alejandro **Cespón Ferriol**

AI/ML Researcher · Applied Machine Learning

Granada, Spain

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Profile

AI/ML researcher and PhD student at the University of Granada, working on explainable AI and large language models. Background in convex optimization, anomaly detection, data analysis and backend development. Co-author of research on distributed outlier detection, decision support and medical image analysis. Interested in applied AI/ML research and engineering roles.

Research & Engineering Experience

DaSCI Institute — *Granada, Spain*

Researcher — *Jan. 2025 - Present*

- Research on explainable artificial intelligence and large language models.
- Institutional profile: DaSCI / Alejandro Cespón Ferriol.

Nitza Development & Pink Zebra Home — *Cuba-USA*

Backend Developer & Data Science — *Jan. 2024 - Apr. 2025*

- Developed REST APIs with Django and Django REST Framework, including JWT-based authentication.
- Processed and analysed datasets with pandas to support operational analysis.

UCLV - Computer Science Department & Research Center — *Villa Clara, Cuba*

Lecturer & Researcher — *Dec. 2023 - Present*

- Taught Discrete Mathematics and Numerical Analysis in the Computer Science degree programme.
- Research on convex optimization, anomaly detection, distance metric learning and distributed algorithms.
- Contributed to AI and pattern-recognition research projects.

UCLV - Artificial Intelligence Research — *Villa Clara, Cuba*

Undergraduate Researcher — *Mar. 2022 - Dec. 2023*

- Researched convex optimization methods for anomaly detection.
- Contributed to a climatic forecasting system for potato late blight.

Technical Skills

ML	Python, PyTorch, scikit-learn, pandas, NumPy, TensorFlow.
LLMs	LangChain, LangGraph, Retrieval-Augmented Generation, Chroma.
Agents	OpenCode coding agent, Model Context Protocol (MCP) integrations, agent skills and harnesses.
Data	SQL, PostgreSQL, SQL Server, Django, REST APIs, Docker, Git.
R&D	Explainable AI, anomaly detection, convex optimization, distributed learning, Latex.

Education

University of Granada — *Granada, Spain*

PhD studies in Information and Communication Technologies — *Nov. 2024 - Present*

Universidad Central "Marta Abreu" de Las Villas — *Cuba*

MSc in Computer Science — *Dec. 2023 - Sep. 2024*

Universidad Central "Marta Abreu" de Las Villas — *Cuba*

BSc in Computer Science — *Sep. 2019 - Dec. 2023*

Selected Certifications

2026 **Robocorp Automation Level I**, Robocorp

2025 **Writing scientific texts with LaTeX and version control with Git**, University of Granada

2024 **Intermediate Machine Learning**, Kaggle

2024 **Intro to Deep Learning**, Kaggle

2024 **Data Visualization**, Kaggle

Selected Publications & Research

Research areas: Explainable AI and Large Language Models; Distributed Optimization and Anomaly Detection; Applied machine learning in healthcare.

Development of a Distributed Outlier Detection Method Based on the Alternating Direction Method of Multipliers

JAMRIS

Distributed outlier detection with ADMM — *2025*

- Alejandro Cespón Ferriol, H. R. González Diez and C. A. Morell Pérez. *JAMRIS*, 19(2), 1–7. DOI: 10.14313/jamris-2025-009.
- Distributed implementation of support vector data description using ADMM, evaluated on multiple datasets.

Heuristic-Boosted Ejection Fraction Estimation from 2D U-Net Segmentation

Springer / WEA 2025

Cardiac function estimation from echocardiography — *2026; online 2025*

- E. D. Serize Portela, A. Rodríguez Sánchez, J. C. Serize Portela, Alejandro Cespón Ferriol and J. I. Ramírez Gómez. CCIS 2702, Part II, pp. 271–282. DOI: 10.1007/978-3-032-08206-0_23.
- Co-authored a study combining U-Net segmentation, geometric volume estimation and regression-based correction.

Intelligent Decision Support System for Intensive Care Units

RCCI

Intelligent decision support for intensive care units — *22 July 2025*

- Alejandro Cespón Ferriol, Ana Isabel Portal, Jorge Alain Caballero Font, Deborah R Galpert Canizares, María Matilde García Lorenzo and Armando Caballero Lopez. *Revista Cubana de Ciencias Informáticas*, 19(2).
- Patient survival classification using hospital data, Random Oversampling and SMOTE; logistic regression, KNN and MLP evaluated with F1, recall and AUC-ROC. The study reports the best results with MLP after Random Oversampling; LIME provides local explanations.

Flujo explicable para clasificar péptidos antimicrobianos con modelo pequeño de lenguaje de proteínas

RCCI

Explainable classification of antimicrobial peptides — *15 October 2026*

- C. Fernández Quintero et al.; co-author: Alejandro Cespón Ferriol. *Revista Cubana de Ciencias Informáticas*, 20(3), 65–82. Article.

A Comparative Study of ProtFlash and ProtT5 Embeddings for Antimicrobial Peptide Classification with Explainable AI

Springer

Protein embeddings for antimicrobial peptide classification — *Online 2026; print 2027*

- M. A. Fernández Marín et al.; co-author: Alejandro Cespón Ferriol. Springer, pp. 142–159. DOI: 10.1007/978-3-032-41324-6_12.

Selected Project Repositories

Intensive Care Hidden Mortality

GitHub

Python, classification, feature attributions, Streamlit, LLM-generated narratives

- Research prototype connecting predictive experiments with an interface for inspecting predictions and explanations.

Distributed outlier detection with ADMM

fed-admm-oc

Federated learning / ADMM

- Federated learning framework for outlier detection based on ADMM. Related repository: I-One-Class-SVM.

AI for enzymes and bioactive peptides

I-IA-AA-Analysis

AI / Bioinformatics

- AI and bioinformatics projects for the discovery of enzymes and bioactive peptides.

UCLV information chatbot

uclv-chatbot

LangChain, Ollama, Chroma, Streamlit

- Demonstration of document chunking, embedding-based retrieval and local language model inference for university information.

Selected Recognition

2023	**Best academic graduate of the university**, UCLV	*Cuba*
2023	**Scientific Merit Award**, UCLV	*Cuba*
2023	**Computer Science Golden Degree**, UCLV	*Cuba*
2019	**Finalist**, ACM-ICPC Cuban Final	*Cuba*
2017	**Member**, National Pre-College Mathematics Team	*Cuba*

Languages

Spanish (native); English (B1 - MCER).